

SECTION 2: INTERNAL AUDITING BENCHMARKS & COMPLIANCE RULES

This section constitutes the Forensic Auditor Rulebook -- a structured compliance framework designed for integration into automated invoice-auditing systems.

AUDIT PILLAR 1: TECHNICIAN ROLE DEFINITIONS

- Apprentice / Helper | Target Rate: \$45-\$65/hr | Authorized Tasks: Basic preventive maintenance, filter replacement, site cleanup, coil washing under supervision. FLAG: Cannot bill for complex diagnostics, refrigerant handling, control programming, or electrical work.
- Journeyman / Service Tech | Target Rate: \$75-\$90/hr | Authorized Tasks: Standard diagnostics, component replacement (capacitors, contactors, motors), refrigerant recovery/recharge, basic electrical troubleshooting. FLAG: Billing above \$90/hr for standard service work without documented specialty justification.
- Master Pipefitter / Lead Installer | Target Rate: \$95-\$115/hr | Authorized Tasks: Complex mechanical installations, welding/brazing large-diameter piping, chiller tear-downs, major retrofits. FLAG: Billing Master-level rates for tasks within Journeyman scope (e.g., filter swaps). FLAG: Billing above \$115/hr without emergency documentation.
- Senior Tech / BAS Controls Specialist | Target Rate: \$110-\$135/hr | Authorized Tasks: Advanced DDC programming, VFD parameter configuration, complex electrical troubleshooting, BAS integration. FLAG: Billing Senior Tech rates for basic thermostat swaps or filter replacements.

AUDIT PILLAR 2: STANDARD PRICING BENCHMARKS (Parts + Labor)

- Standard Filter Replacement: \$50-\$100 per unit. FLAG: Any filter service exceeding \$100 per unit. (Exception: HEPA/MERV-15+ documented with part number).
- Preventive Maintenance Visit: \$120-\$250 per system. FLAG: PM visits exceeding \$250 per standard RTU or split system.
- Capacitor Replacement: \$150-\$300 total. FLAG: Single-run or dual-run capacitor replacements exceeding \$300.
- Refrigerant Recharge (R-410A): \$45-\$85 per lb. FLAG: Exceeding \$85/lb.
- Refrigerant Recharge (R-22): \$100-\$175 per lb. FLAG: Exceeding \$175/lb.
- Contactor Replacement: \$100-\$200 total. FLAG: Exceeding \$200 without documented specialty part.
- Thermostat Replacement (Basic): \$75-\$175 total. FLAG: Basic non-communicating thermostat swap exceeding \$175.
- Blower Motor Replacement: \$350-\$700 total. FLAG: Standard PSC blower motor replacement exceeding \$700.

AUDIT PILLAR 3: LABOR RATE STANDARDS & CRITICAL COMPLIANCE CAPS

- The maximum allowable standard hourly labor rate is strictly \$95/hr for Journeyman-level service work. Any standard invoice billing higher without a documented specialty role must be automatically flagged.
- Higher labor rates of \$120-\$150+/hr are ONLY justified if the invoice explicitly documents an emergency dispatch outside standard business hours (8:00 AM - 5:00 PM, Monday-Friday). Weekend/holiday dispatches qualify. Invoice must explicitly state dispatch/arrival times and use language like "Emergency Call" or "After-Hours".
- Standard diagnostic/trip fees must be capped at \$85-\$120. Emergency after-hours diagnostic fees may not exceed \$180.

AUDIT PILLAR 4: COMMON BILLING IRREGULARITIES (Training Data)

- Category A - Vague/Lump-Sum Billing (FLAGGED):
 - * Example: "Fixed AC - \$600" -> FLAG (No breakdown of parts vs labor, no hours, single-line total).
 - * Example: "Performed repairs to rooftop unit and added freon - \$1,250" -> FLAG (Combines labor and refrigerant, no lbs documented, no hours).
 - * Example: "Replaced bad parts in condenser - \$450" -> FLAG (No specific parts named, no part numbers).
- Category B - Properly Itemized Billing (APPROVED):
 - * Example: "Replaced 45/5 MFD Dual Run Capacitor (Part #CAP-45-5-440): 1 EA @ \$45.00. Labor: 1.5 Hrs @ \$85/hr. Total: \$172.50" -> PASS.

* Example: "Leak search and repair on RTU-2. Recovered and recharged 4 lbs R-410A @ \$65/lb. Labor: 2.0 Hrs @ \$90/hr. Total: \$440.00" -> PASS.

AUDIT PILLAR 5: TASK-TO-ROLE MAPPING RULES

- Filter Replacement / Coil Washing / Drain Pan Cleaning: Capped at \$65/hr (Apprentice/Helper). FLAG if tech bills >\$65/hr.
- Thermostat Swap (Basic): Capped at \$90/hr (Journeyman). FLAG if BAS Specialist or Master Tech bills premium rates.
- Capacitor/Contactor/Fuse Replacement: Capped at \$90/hr (Journeyman). FLAG if Master or Senior Tech bills premium rates for basic component swap.
- Compressor Replacement: Capped at \$115/hr (Journeyman/Master).
- VFD Parameter Programming / BAS DDC Configuration: Capped at \$135/hr (BAS Controls Specialist).